

INDIAN INSTITUTE OF TECHNOLOGY JAMMU - 181221

B.Tech. Electrical, Mechanical and Materials Engineering

END SEMESTER EXAMINATION - Date: 22.04.2026

MT-1-02(TO) - ENGINEERING STANDARDS

Time: 120 Minutes

Maximum Marks: 50

Answer all the questions, each question carries 2 marks

1. Compare and contrast "*waiver*" and "*tailoring*" of standards.
2. How the potential conflict between a "patented product" and "standards" can be addressed.
3. Compare and contrast *de-facto* and *de-jure* Standards.
4. Briefly explain Innovation/standardization cycle.
5. Compare and contrast validation and verification process in relation to product development as per IEEE guidelines.
6. Comment on the importance of national building codes and electrical codes, with a suitable example.
7. Justify with an example, how standards can be helpful for promotion of businesses.
8. Give any one example for a "systems" standard. Explain Briefly.
9. Compare and contrast jurisdictional and non-jurisdictional standard.
10. List any four accredited agencies for the development of voluntary consensus standards.

Answer all the questions, each question carries 5 marks

11. A manufacturer intends to produce "*stents*", a medical device for enlarging of narrowed blood vessels. What are the key items that must be considered when developing the conformity assessment sheet regarding (1) Product functionality (2) Manufacturing details (3) Materials used for the device and (4) Clinical trial documentation and certifications?
12. Choose a product/service of your interest, **ANY ONE**, from the below mentioned list and develop a standard with important terminologies, scope, general requirements, testing methods, mandatory specifications, labeling and miscellaneous, if any.

(A) UPI – Unified payment systems
(B) Non-destructive testing
(C) Drones for logistics
13. What are the various **FACTORS** that will lead to non-conformity of products? List all the approaches that a manufacturer can follow in dealing with non-conformity of products. Discuss the advantages and disadvantages of each approach with suitable examples.

14. Discuss how standards can influence innovation, both in a positive and a negative manner with a suitable example. (*Hint: prescriptive vs. performance*).
15. An automobile manufacturer wishes to develop and produce a "**self-driving vehicle**" for use in public roads. Describe how different types of standards directly or indirectly impact the development with regards to mechanical, safety, sensors and electrical requirements.
16. Enumerate the role of standardization and conformity marking in enhancing free-trade among member countries of EU.